

Omega-S: A Functional Resilience Index for LLM Fine-Tuning

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Abstract

Fine-tuning a large language model on new data degrades what it previously learned. We present **Omega-S**, a drop-in penalty computed from the weight matrix alone: it needs no previous-task data, no Fisher matrix and no stored copy of the old weights, which is what the standard baseline for this problem requires. It is three lines in an existing training loop and adds under 4% to the cost of a step.

Retention. On Llama-3-8B with LoRA, fine-tuned from code to prose and measured by HumanEval over ten seeds, Omega-S leaves the model with more of its original capability than no regularisation on **9 of 10 seeds** ($0.173 \rightarrow 0.238$ absolute pass@1, a 37.7% relative increase; exact sign test, one-sided $p = 0.011$, Wilcoxon $p = 0.006$). As a retention ratio the same comparison gives $62.9\% \rightarrow 84.1\%$. It also beats tuned weight decay on 10 of 10 seeds ($p = 0.002$) and tuned EWC on 8 of 10 ($p = 0.014$) on the same absolute measure, with every arm re-measured in the same session so that the comparison does not mix effect with run-to-run variation.

The design space. Turning the index into a penalty leaves three choices open: how the graph is built from the weights, which form of the objective is used, and how the modularity factor is oriented. We enumerate all three and report what each costs, including a contrast-preserving construction that does what it was designed to do and makes retention *worse* on all ten seeds. Two of the three choices turn out not to matter for the mechanism, and we show why.

Mechanism, measured rather than asserted. Omega-S is topological by construction, its objective being built from $\text{Tr}(A^3)$, but we measured which of its four factors actually moves during training and three of them do not. Their elasticity with respect to the weights is at or below 10^{-4} , against 9×10^{-3} for the degree-variance term, before and after training alike; during a run the degree variance falls by 5.9% in median while the other three stay under 0.1%. As implemented, the composite objective reduces in practice to a penalty on the variance of node degrees, a quantity that means row magnitude in square modules and directional alignment in non-square ones. An ablation shows the alignment channel leading the magnitude channel on 8 of 10 seeds. We also built the contrast-preserving reformulation that earlier drafts proposed as an open direction, and report that it makes retention worse on all ten seeds under either orientation: the inert clustering channel was not a defect holding the method back.

How much run-to-run variation this setting carries. Repeating an identical configuration, same seed and same hardware, gives a standard deviation of 0.104 in retention ratio. We report it because we have not found it quantified for low-rank fine-tuning of language models, and because it bounds the reading of any seed-paired comparison, including ours.

Two forms of the penalty exist and they are not interchangeable; they rank oppositely across the two settings above, and we define both explicitly (Section 2.1). Code, per-run results and the full record of negative results are at <https://github.com/BiomeMakers/OmegaS-LLM> under AGPL-3.0 for research use.

1. Introduction

1.1. An index from soil, and what it measures

The quantity this paper minimises did not originate in machine learning. It comes from a body of work on agricultural microbiomes, where the object of study is a co-occurrence network of taxa and the practical question is whether a soil community is functionally integrated or dominated by a few organisms. A survey of 350 vineyard soils across two countries [1] found that the communities associated with resistance to perturbation share a signature: *higher* clustering, *lower* modularity and *lower* co-exclusion. Communities with the opposite signature are diverse by count and fragile in practice, because a small set of organisms carries the connectivity.

That signature was formalised as an index, $\Omega = C \cdot D / (M \cdot \text{Coex})$, on four measures of a weighted graph: clustering C , density D , a modularity measure M , and a concentration term Coex given by the variance of the degree sequence. The index is high when connectivity is distributed and low when it is monopolised by a few nodes. Its theoretical development, including the identity $\text{Tr}(A^3) = \sum_i \lambda_i^3$ and the conditions under which the index is comparable across graphs, is in a companion manuscript [21].

1.2. The Production Problem: Weight Monopolies

Every organisation that fine-tunes large language models on proprietary data faces the same hidden cost: the model forgets. Update a general-purpose model on customer support data, and it loses coding ability. Fine-tune on legal documents, and general reasoning degrades. This is catastrophic forgetting, and it means that the compute cost of fine-tuning is paid twice: once to train, and once to repair what broke.

Standard regularisers do not solve this. Weight decay pushes all parameters toward zero indiscriminately. SAM optimises loss landscape curvature. Neither addresses the structural root cause: during training, a small subset of neurons progressively concentrates disproportionate connectivity across layers, creating *weight monopolies*. When new data arrives, these monopolies are overwritten first, destroying the representations that made the model valuable.

Omega-S targets weight monopolies directly.

1.3. Our Contributions

A regulariser from an ecological index. We define Omega-S, a penalty on the connectivity graph $WW^\top$ of a weight matrix, built from $\text{Tr}(A^3)$ and estimated by Hutchinson’s method so that its cost is $\mathcal{O}(N^2)$ rather than $\mathcal{O}(N^3)$. It needs no previous-task data, no Fisher matrix and no stored copy of the old weights, and adds under 4% to the cost of a training step.

A design space, mapped rather than assumed. Three choices are open in turning the index into a penalty: the graph construction, the form of the objective and the orientation of the modularity factor. We screen six constructions, define both forms of the objective, and settle the orientation from the ecological observation the index formalises. The variant that looks most promising a priori, a contrast-preserving construction that revives the clustering term, is measured and makes retention worse on all ten seeds.

Retention on Llama-3-8B. The selected variant leaves more code capability after a sequential fine-tune than no regularisation on 9 of 10 seeds ($0.173 \rightarrow 0.238$ absolute HumanEval, +37.7% relative, sign test $p = 0.011$, Wilcoxon $p = 0.006$), and beats tuned weight decay on 10 of 10 and tuned EWC on 8 of 10 on the same measure, with every arm re-measured in one session.

A mechanism that contradicts the name. Omega-S is topological by construction, and we measure that three of its four factors are numerically inert: elasticities at or below 10^{-4} against

9×10^{-3} for degree variance, before and after training alike, with the degree variance the only factor that moves during a run. As implemented, the objective reduces in practice to a penalty on the variance of node degrees. We report this because a method whose name promises one thing and whose gradient does another should say so.

How much run-to-run variation this setting carries. Repeating an identical configuration gives a standard deviation of 0.104 in retention ratio. We have not found this quantified for low-rank fine-tuning of language models, and it bounds the reading of every seed-paired comparison in this literature, ours included.

2. Method

2.1. The Omega-S Regularizer: two forms

Omega-S exists in two forms, and the distinction matters enough that we define both. The first is the direct penalty we started from; the second is the log-ratio form that the experiments in this paper evaluate, and which is what the reported results refer to. We give the reasoning that leads from one to the other, because it is the same reasoning that later explains why the clustering channel is inert (Section 3.3).

Form 1: the raw trace penalty. For a weight matrix $W \in \mathbb{R}^{m \times n}$, define the pseudo-topological adjacency matrix $A = WW^\top \in \mathbb{R}^{m \times m}$ and the per-layer penalty

$$\Omega_\ell^{\text{raw}} = \frac{1}{\text{numel}(W)} \cdot \widehat{\text{Tr}}(A^3), \quad (1)$$

where $\widehat{\text{Tr}}(A^3)$ is the Hutchinson estimate with n_p Rademacher probe vectors. This is the form used in our earliest experiments and it is the one whose spectral interpretation is most direct: $\text{Tr}(A^3) = \sum_i \lambda_i^3$ exactly.

It has a defect that is visible from the algebra alone. Since A is quadratic in W , Eq. 1 is homogeneous of degree six in W , so its gradient is homogeneous of degree five. The force the penalty exerts therefore scales as the fifth power of the weight magnitude: it explodes when weights grow and switches itself off when they shrink. In a run where any other pressure reduces weight magnitude, and structured sparsity does exactly that, the penalty quietly stops acting. This is not a hypothesis about why the raw form underperforms; it is a property of the functional, and it is consistent with the raw arm being the weakest in the ten-seed sweep (Section 3.2).

Form 2: the log-ratio composite. The remedy is to build the penalty from quantities that do not inherit the scale of W , and to take logarithms so that the gradient is normalised by the value rather than proportional to it. Concretely, let

$$W_{\text{corr}} = \begin{cases} WW^\top & \text{if } m \neq n, \\ W & \text{if } m = n, \end{cases} \quad S = \sigma(|W_{\text{corr}}|), \quad A = \frac{1}{2} (S + S^\top), \quad (2)$$

with σ the logistic function. We state the case distinction explicitly because the reference implementation contains it and the runs reported here therefore used both branches, one per module type; Section 3.6 gives the measured consequence and our reading of it.

From A we form four quantities. Writing $k = A\mathbf{1}$ for the degree sequence and $L = \text{diag}(k) - A$ for the graph Laplacian,

$$\begin{aligned} D &= \text{mean}(A), & \text{Coex} &= \text{var}(k) + \varepsilon, \\ C &= \frac{\widehat{\text{Tr}}(A^3)}{\|A\|_F^3 + \varepsilon} + \varepsilon, & M &= |v^\top L v| + \varepsilon, \end{aligned} \quad (3)$$

where v is obtained by three steps of shifted power iteration on $2k_{\max}I - L$, mean-centred and renormalised at each step. The mean-centring deflates the trivial all-ones eigenvector, so M estimates the algebraic connectivity, that is the Fiedler value $\lambda_2(L)$, rather than modularity in the community-detection sense; we retain the symbol M for continuity with the index it derives from, and note that minimising λ_2 is what pushes the graph towards a more modular structure. Penalising the Fiedler value of a network’s weight graph during training was introduced by Tam and Dunson [6] as Fiedler regularisation, and our M factor coincides with theirs in both quantity and direction; what differs here is that λ_2 enters as one factor of a composite objective alongside a triadic term, and, as Section 3.3 reports, it is not the factor that carries the effect we measure. The per-layer penalty is

$$\Omega_\ell^{\text{lib}} = \log\left(\frac{M \cdot \text{Coex}}{C \cdot D + \varepsilon}\right). \quad (4)$$

Minimising Eq. 4 therefore lowers M and Coex and raises C and D : it pushes the weight graph towards lower degree variance and lower algebraic connectivity, and towards higher clustering and density. In either form the total penalty added to the training loss is

$$\Omega = \lambda_\Omega \sum_\ell \Omega_\ell, \quad \mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \Omega, \quad (5)$$

applied every K optimisation steps, with λ_Ω calibrated at run time so that the penalty gradient is a fixed fraction of the task gradient.

Which form produced which result. The two forms are not interchangeable and the results in this paper do not all come from the same one. The retention experiments on Llama-3-8B (Section 3.2), which are the main claim, use the log-ratio composite of Eq. 4. Earlier structural-pruning experiments on an MLP, not reported here, predate that formulation and use the raw penalty of Eq. 1. We make the mapping explicit because “Omega-S” otherwise names two objectives, and because one of them is the arm that performs worst on retention.

The fix and its cost. Two features of Eq. 2 and Eq. 4 remove the scale sensitivity of the raw form. The logistic maps every entry into $[0.5, 1)$ regardless of the magnitude of W , so A no longer inherits the scale of the weights at all; and the logarithm makes the gradient proportional to $\nabla f/f$ rather than to ∇f , which removes the remaining dependence on the size of the penalty value. Empirically this is the difference between the weakest arm in the sweep and the strongest (Section 3.2).

It is worth stating plainly that the same choice carries a cost we did not anticipate when we made it, and that Section 3.3 measures. A bounded map applied to a quantity with a very large dynamic range does not merely rescale it, it collapses it: with $|W_{\text{corr}}|$ spanning several orders of magnitude, σ sends almost every entry to the same value, and the clustering term C , which depends only on the contrast between entries, is pinned near a constant. The map that solved the scale problem is the map that disabled the topological channel. We report both halves of that trade because only one of them was visible when the design was fixed.

2.2. Computational Efficiency

For typical transformer dimensions ($m = 4096$, $n_p = 3$), the sequential decomposition yields approximately 79% reduction in computation time relative to the explicit A^3 method. The computation is purely structural and requires no gradient graph traversal.

When applied to LoRA adapters (rank $r = 8$), the weight matrices are small enough that Omega-S requires **zero inter-GPU communication** in FSDP environments, as the adapter weights are not sharded.

2.3. Application Frequency

We recommend applying Omega-S every $K \in [10, 50]$ steps. The topological impact is cumulative and does not require per-step application. At $K = 10$, amortised overhead is below 4% in all tested configurations.

Quick Start

Omega-S is designed as a drop-in addition to any existing PyTorch training loop. Adding it requires three lines of code:

```
from omega_s import StochasticOmegaS

omega = StochasticOmegaS(lambda_omega=0.05, k=10, n_probes=3)

# Inside your training loop:
loss = model(inputs, labels=inputs).loss
loss = loss + omega(model)  # add topological penalty every K steps
loss.backward()

For LoRA fine-tuning with HuggingFace + PEFT (recommended):

# Works with any PEFT model: zero changes to your existing setup
model = get_peft_model(base_model, lora_config)
omega = StochasticOmegaS(lambda_omega=0.05, k=10)

for step, batch in enumerate(dataloader):
    loss = model(**batch).loss + omega(model)
    loss.backward()
    optimizer.step()
```

Full examples, FSDP integration, and reproduction scripts for all experiments in this paper are available at: <https://github.com/BiomeMakers/OmegaS-LLM>

For commercial licensing or research partnerships: acedo@biomemakers.com

2.4. The design space, and which variant we use

Turning the index into a training penalty leaves three choices open, and each admits more than one reasonable answer. We enumerate them here and report what each costs, because the choices are not obvious a priori and because two of the three turn out not to matter.

How the graph is built from the weights. The index is defined on a weighted graph and a weight matrix is not one, so a construction is needed. The default is a bounded map on the Gram matrix, $A = \sigma(|WW^\top|)$. Six alternatives were screened statically on the base weights of the model: the bounded map on the raw matrix, a row-wise cosine affinity, a rank transform, a thresholded graph, a doubly stochastic scaling and a local scaling. The screen measures the *triadic excess* C/D , which equals one on a graph with no triadic organisation over its density and is therefore an absolute reference that C alone does not provide.

The default construction returns $C/D = 1.0000$ to four decimal places in every module tested, equal to its permutation null: the graph it builds carries no triadic content over its density at all. The cosine construction lifts that to a median of 1.36 and makes the clustering term move. That looks like the obvious improvement, and it is not: trained on the ten-seed protocol, the cosine composite loses on all ten seeds (0.547 against 0.841, Wilcoxon $p = 0.002$). The

ordering is monotone in how much the clustering channel moves, which is the opposite of what a topological reading would predict. We keep the default construction and report the negative result in Appendix B.12.

Which form of the objective. The index can enter the loss as the raw triadic trace $\text{Tr}(A^3)$ or as the log-ratio of the four factors, $\log(M \cdot \text{Coex}/(C \cdot D))$. These are not interchangeable and they rank oppositely across our two settings: the composite form wins on retention; in separate experiments on structured pruning, not reported here, the two rank oppositely. We define both explicitly rather than presenting one as *the* method, and use the composite throughout.

How the modularity factor is oriented. M enters the denominator of the index, so whether it is taken as a modularity measure or as its inverse changes the direction in which the penalty pushes. The ecological observation that motivated the index fixes this: resistant communities show *lower* modularity, so Ω must decrease with modularity, so M must be a modularity measure. A graph that separates easily into modules has small algebraic connectivity λ_2 , which makes $1/\lambda_2$ the quantity that is large on modular graphs. We therefore take $M = 1/\lambda_2$.

We also ran the opposite orientation, $M = \lambda_2$, over the same ten seeds. It gives 76.6% retention against 84.1%, losing on 8 of 10 seeds. The difference, +0.075, is smaller than the run-to-run standard deviation of 0.104 we measure in Section 3.4, so the two orientations are *not distinguishable from each other* at ten seeds; both are distinguishable from the baselines. We report the comparison because of what it says about mechanism rather than about performance: Section 3.3 shows that the modularity term is numerically inert under either orientation, so any difference between them cannot be a topological effect of M . What inverting the term does change is the *scale* of the objective, from -1.06 to -10.73 on a representative matrix, and since the penalty coefficient is calibrated by gradient ratio, a different scale places the one live factor at a different operating point. The control is that applying the same inversion under the cosine construction, where the objective does not jump in scale, changes nothing at all: 4 of 10 seeds, mean difference +0.010, $p = 0.83$.

What this leaves. Of the three choices, one is settled by the domain the index comes from, one is settled empirically against a negative result, and the third is a genuine fork with two answers that serve different purposes. The configuration used from here on is: bounded map on the Gram matrix, log-ratio form, $M = 1/\lambda_2$, penalty applied every ten steps with the coefficient calibrated so the penalty gradient is 3% of the task gradient.

3. Experiments

3.1. Experimental Setup

Base experiments use a three-layer MLP trained on Split-MNIST (digits 0–4) with architecture [784→256→128→10], Adam optimiser ($\text{lr} = 10^{-3}$), batch size 128, 5 epochs, seed 42.

LLM experiments use Llama-3-8B [15] with LoRA adapters ($r=8$, $\alpha=16$). The adapter is inserted into **q_proj** and **v_proj** of every attention layer, which is the placement the original LoRA work selects by ablation [9] and the one in near-universal use since; we use it here so that the comparison against baselines that assume it is like for like.

Omega-S hyperparameters differ between the two runs, and we give both because the repository quick-start reproduces the first and not the second. The five-seed run of Section B.13 uses a fixed $\lambda_\Omega = 0.05$ with $K = 10$ and $n_p = 3$ probe vectors. The ten-seed run of Section 3.2 uses $K = 10$, $n_p = 16$ probes (three was found to give an estimator dominated by sampling noise), a sampled subset of eight modules per application, and a λ_Ω calibrated at run time so that the

penalty gradient is a fixed fraction 0.03 of the task gradient rather than a fixed constant; the resulting λ_Ω therefore varies by seed, between 16 and 41 in this run.

3.2. Stability–Plasticity Trade-off

We evaluate Omega-S in a sequential fine-tuning setting using Llama-3-8B with LoRA adapters ($r=8$, $\alpha=16$), transferring from code (CodeAlpaca-20k [17], 5,000 samples) to prose (Wikitext-2 [18], 5,000 samples). Knowledge retention is measured via HumanEval pass@1 [16] (164 problems) evaluated before fine-tuning, after Domain A (code), and after Domain B (prose). The experiment is replicated with five independent random seeds to assess robustness, with the Omega penalty verified to be connected to the autograd graph before each run.

Table 1: **Main result (ten-seed experiment)** on Llama-3-8B (LoRA, sequential fine-tuning code $\rightarrow$ prose), ten random seeds, penalty connected to the autograd graph, per-arm dedicated RNG. *Code kept* is absolute HumanEval pass@1 after the prose task and is our primary metric; it is not a measure of plasticity, since the second task is prose and HumanEval measures code throughout. *Retention* is the same quantity divided by post-code HumanEval; values above 1.0 mean net positive transfer on that seed. All arms use the hyperparameters selected in Section B.11; none was re-tuned for this table, which would have given the baselines a best-of-several advantage that Omega-S did not receive. Every arm in this table was measured in the same session, on the same hardware, so that no comparison mixes the effect with the run-to-run variation quantified in Section 3.4. The *wins* column counts seeds on which Omega-S keeps more code than that arm, on the absolute measure, and gives the exact sign test and the Wilcoxon signed-rank test on the paired differences.

Arm	Code kept	Retention	Std	Omega-S wins	p (signs / Wilcoxon)
No regulariser	0.173	62.9%	0.144	9/10	0.011 / 0.006
Weight decay	0.174	61.9%	0.090	10/10	0.001 / 0.002
EWC ($\lambda=10^3$, swept)	0.190	69.1%	0.185	8/10	0.055 / 0.014
Row-norm control	0.030	17.5%	0.093	10/10	0.001 / 0.002
Cosine composite	0.156	54.7%	0.125	10/10	0.001 / 0.002
Omega-S (log-ratio)	0.238	84.1%	0.128	—	—

Table 2: **The two orientations of the modularity factor**, ten seeds each, same hardware and same session. The framework defines M as a modularity measure and therefore places its inverse in the objective; the reference implementation estimated algebraic connectivity, the inverse quantity. The corrected orientation is the better of the two, but the paired difference between them, +0.075, is smaller than the run-to-run standard deviation of 0.104 measured in Section 3.4, so *the two orientations are not distinguishable from each other* on ten seeds. What is distinguishable is each against the baselines, and those comparisons are in Table 1.

Orientation	Retention	Std	vs. the other
As implemented ($M \approx \lambda_2$)	76.6%	0.145	2/10
As defined ($M \approx 1/\lambda_2$)	84.1%	0.128	8/10, $p = 0.055$

(Table 11 is reported in the appendix.)

In the ten-seed experiment, the log-ratio Omega-S improves retention over *tuned* weight decay on 10 of 10 seeds (mean 61.9% $\rightarrow$ 84.1%, +17.3 pp) and over *tuned* EWC on 8 of 10 (+17.3 pp), and it does so while ending with the highest absolute code capability of any arm (HumanEval pass@1 = 0.238 after the prose task, against 0.173 with no regulariser). We are careful about what that second number is. It is the numerator of the retention ratio reported in absolute terms, and it is the quantity a practitioner cares about, namely how much of the original capability

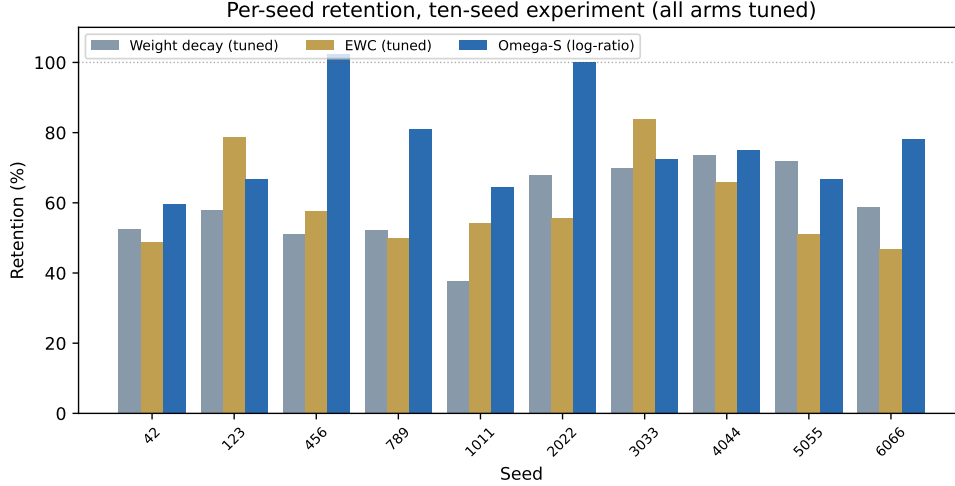


Figure 1: Per-seed retention in the ten-seed experiment, all arms tuned. Omega-S (right bars) exceeds both baselines on most seeds; the two seeds where it trails EWC (123, 3033) are those where EWC itself is highest. The dotted line marks retention = 100% (full retention of the code capability after the prose task).

survives. It is *not* a measure of plasticity, and we do not describe it as one: the retention metric is HumanEval throughout while the second task is prose, so nothing in this experiment measures performance on the new task. Whether Omega-S costs plasticity on the second task is untested here and would require a Wikitext perplexity measurement we did not make.

On the absolute measure Omega-S beats the no-regulariser arm on 9 of 10 seeds (exact sign test, one-sided $p = 0.011$), one more than the 8 of 10 the ratio gives. The discrepancy is instructive. On seed 5055 Omega-S ends with more code capability in absolute terms (0.207 against 0.171) and yet a lower retention ratio, because it had also reached a higher post-code baseline. The ratio penalises an arm for doing better on the first task, which is the main reason we lead with the absolute figure.

The penalty is applied during both tasks, so post-code HumanEval is arm-specific by construction rather than a shared baseline; this is why we lead with the absolute metric. The two are nonetheless statistically indistinguishable between the Omega-S and no-regulariser arms: the means are 0.291 against 0.268, Omega-S is ahead on 5 of 10 seeds with one tie (exact sign test, one-sided $p = 0.50$), and the mean paired difference of +0.023 sits well inside its own spread across seeds (sd = 0.050). The retention gain therefore does not arise from a different starting point.

We ran this experiment after a first five-seed run (Table 8) that showed the same direction more weakly (4/5 over weight decay); the effect strengthened, rather than regressing to the mean, when we doubled the seeds and tuned every arm on the same footing.

This comparison against EWC is worth stating in full, because Omega-S wins it while asking for strictly less. EWC is the standard method for catastrophic forgetting, and to build its penalty it must retain a copy of the previous-task weights and estimate a Fisher information matrix by passing *previous-task data* back through the model, so it requires continued access to the old data, which in production is often unavailable (privacy, storage, data that has expired). Omega-S reads only the current weights: no stored optimum, no Fisher matrix, no previous-task data. It therefore beats, on 8 of 10 seeds, the field’s standard forgetting baseline while removing that baseline’s core operational requirement. (We do not claim lower per-step compute: Omega-S’s penalty carries a Hutchinson estimate over WW^T whose cost we have not benchmarked against

EWC’s; the advantage we do claim is operational (no dependence on prior data), not raw speed.)

The margin is uneven, and its shape suggests a ceiling. The per-seed advantage is far from uniform: against the unregularised arm it ranges from -4.4 to $+54.3$ percentage points. An earlier version of this work read that spread as evidence of two regimes, topological on some weight graphs and closer to norm-balancing on others, set by each seed’s initialisation. We tested that directly by measuring the four structural components on every seed (Section 3.3) and it is not supported: on all ten seeds the penalty moves the same component, the variance of node degrees, and the other three do not move at all. We record the two-regime hypothesis as refuted.

What the per-seed data do show is simpler. The gain over the unregularised model shrinks as that model’s own retention rises, with a rank correlation of $\rho = -0.73$ across the ten seeds, and the single seed where Omega-S loses to that baseline (42) is one where the baseline already retains two thirds of its capability. The margin has the shape of a ceiling effect: the penalty helps roughly in proportion to how much forgetting there is to prevent. We report this as a pattern and not as a finding, because at $n = 10$ separating a genuine ceiling from regression to the mean would need more seeds than we ran.

Not equivalent to norm-balancing. A recurring concern for any penalty on $WW^\top$ is that it may reduce, in practice, to equalising the norms of the weight rows, which would make it a reparametrisation of weight decay rather than a topological criterion. Table 1 includes the direct control: a penalty on the variance of row norms, at the same calibration target, run over the same ten seeds in the same session. It reaches 17.5% retention against 84.1% for Omega-S, losing on all ten seeds ($p = 0.002$). Row-norm equalisation on its own does not buy retention in this setting.

That control is run at Omega-S’s calibration target rather than its own, which is a deliberate but debatable choice: it tests whether the *same* penalty budget spent on norms achieves what spending it on degree variance does. A version tuned on its own grid, and a protocol defect we found in the original selection, are in Appendix A.

3.3. What the penalty actually moves: a direct measurement

Omega-S is *by mathematical definition* a topological regulariser: its objective is built from $\text{Tr}(A^3)$, the third spectral moment of the weight graph $A = WW^\top$, whose node-wise decomposition carries information no purely spectral index determines (see Method). That is a statement about the functional. It leaves open a separate, empirical question: under the training conditions of these experiments, *which* of the objective’s components does the penalty actually move to produce the retention gain? We measured this directly rather than assuming it.

For each of the ten seeds we trained task A twice, once with no regulariser and once with Omega-S, and recorded the four components of the objective (C clustering, D density, M modularity, Coex degree variance) on the effective LoRA weights, together with the retention gain of that seed. The result is unambiguous and one-sided. The change in clustering is $\Delta C \approx 0$ on every seed (e.g. seed 456: C moves from 0.999708 to 0.999702), whereas degree variance falls substantially (Coex drops 5–10%, e.g. seed 456: $201.6 \rightarrow 182.8$). In other words, the retention effect is carried entirely by the denominator’s Coex term, not by the numerator’s clustering term.

The reason is structural, not incidental. The pseudo-adjacency is built as $A = \sigma(|WW^\top|)$, and the logistic σ compresses all entries into $[0.5, 1)$; A is therefore close to an all-ones matrix, on which the normalised clustering $\text{Tr}(A^3)/\|A\|_F^3$ is pinned near its ceiling ($C \approx 0.9997$) regardless of the underlying weight structure. The clustering channel is saturated and inert *by construction*: Omega-S cannot raise C because σ has already removed the contrast that would let it. What remains free to move is Coex, the variance of node degrees. We name this plainly as a design

shortcoming rather than a technical detail: the logistic map in the construction of A is not motivated by the topological objective, and in retrospect it is precisely what disables the clustering channel the method is built to exploit. A construction that preserves contrast (Appendix A) would be preferable in any future formulation; we report the saturation here, having measured it, rather than leaving it for a reader to infer.

How much of the objective is actually live: a direct measurement of all four factors.

The statement above concerns the clustering term, and a reader is entitled to ask what the other three do. We measured the *elasticity* of each factor with respect to the weights, that is the relative change in the factor produced by a relative change in W along a fixed random direction. The quantity is dimensionless, so it is comparable across factors and modules, and a value of zero means the factor cannot contribute gradient at all. Measured on the base weights of Llama-3-8B over ten attention projections spanning layers 0 to 31, each with the branch the implementation actually applies to it:

	C	D	M	Coex
Elasticity, base weights	0.0000	0.0000	0.0001	0.0091
Elasticity, trained weights	0.0000	0.0000	0.0001	0.0139
Relative change during a run	−0.00%	+0.07%	+0.06%	−5.87%

The three rows measure three different things and agree. The first is the sensitivity of each factor on the model’s base weights; the second is the same quantity after a full fine-tune, which tests whether a factor that looks inert at initialisation comes alive once the weights move; the third is the actual change in each factor over a run, relative to the unregularised arm, median over the ten seeds.

Three of the four factors are numerically inert on all three measures, and one carries everything. Degree variance responds roughly ninety times more strongly than the modularity term, is the only factor above the noise floor, and is the only one that moves during training, falling 5.87% in median while the other three stay under 0.1%. **As implemented and in these conditions, the composite objective reduces in practice to a penalty on the variance of node degrees.**

The middle row answers a question the first row alone leaves open. If a factor looks inert at initialisation, it may still come alive once the weights move, and the two orientations of M do differ in outcome (Section 2.4), which would be explained if M started contributing during training. It does not: the elasticity of M is 0.0001 before and after training alike, and its relative change over a run is +0.06%, indistinguishable from the clustering and density terms. Whatever separates the two orientations, it is not the modularity term acting on the weights, which is what leaves the operating-point account of Section 2.4 as the one that fits.

The result holds under both orientations of M : the numbers above are measured with $M = 1/\lambda_2$, and the degree-variance change under the other orientation was −5 to −10% with the other three factors equally inert. The mechanism does not depend on which of the two is used, which is one more reason to read the difference between them as a change of operating point rather than as a topological effect.

It is worth being precise about what that inertia is a property of, because the natural reading is that the framework’s factors are weak and that is not what the measurement says. The bounded map is what flattens them: in the ecological setting from which the index derives, modularity is among the strongest signals available, separating management types with $r^2 = 0.955$ in the survey of soil communities that motivated the construction, comparable to the co-exclusion term and higher than clustering. A factor that carries most of the signal in the domain of origin and none under a saturating map applied to neural weights is telling us about the map, not about the factor. The same holds for the orientation question above: the direction of M is fixed in the

framework by that ecological observation and not by convention, and it is only here, where the term does not move, that the orientation ceases to matter.

We state the consequence plainly. **As currently formulated, and in these conditions, Omega-S operates by reducing the variance of node degrees in the weight graph**, a quantity that is strongly, though not perfectly, correlated with the variance of weight-row norms (we measure $r \approx 0.60$ here, leaving roughly 40% of degree-variance unexplained by row norms). It is therefore not, in practice, a clustering-driven topological method, even though its objective is topological in form. We found no evidence of a second, clustering-driven regime: across all ten seeds the movement is dominated by Coex, so the bimodal per-seed pattern of Table 11 is not explained by two structural mechanisms in these data.

Three points follow, and we are careful not to overclaim in either direction. First, this does *not* weaken the empirical result: Omega-S still leaves more code capability than no regularisation on 9/10 seeds ($p = 0.011$), and leads weight decay on 9/10 and EWC on 8/10 in retention, without access to previous-task data. The method’s demonstrated utility stands; only the explanation of *why* it works is revised, from “clustering/topology” to “degree-variance control”. Second, degree-variance control is itself a legitimate and, to our knowledge, uncharacterised lever for LoRA continual learning, distinct from generic weight decay; the row-norm control, now swept on its own grid, does not reliably beat no regularisation (6/10) while Omega-S exceeds it on 8/10, which bounds the $\approx 40\%$ unexplained component without fully quantifying it. Third, and this is a direction we offer to the community rather than a result we claim here: the topological channel is inert only because of the saturating σ . A reformulation that keeps C informative, for instance replacing $\sigma(|WW^\top|)$ with a contrast- preserving map such as a temperature-scaled or rank-normalised affinity, or penalising $\text{Tr}(A^3)$ on a thresholded graph so that clustering is not pinned near 1, would let a genuinely clustering-driven variant of Omega-S be tested. Whether such a variant outperforms the degree-variance mechanism measured here is untested. We sketch this reformulation in Appendix A and leave its evaluation to future work.

What the degree sequence measures, and a construction detail we must declare.

Two further measurements refine the statement above. The first is a discrepancy between the text and the reference implementation that we report rather than leave for a reader to find. The implementation forms the pseudo-adjacency as $\sigma(|WW^\top|)$ only when W is non-square; when W is square it applies $\sigma(|W|)$ elementwise, without forming the Gram matrix. In Llama-3-8B, `q_proj` is square and `v_proj` is not, so the runs reported here used one module of each kind and therefore two constructions at once, while the description above and in the Method section covers only the Gram form. The measured consequence is not cosmetic: on `q_proj` the degree-variance term differs by a factor of roughly 400 between the two branches. This does not affect the reported numbers, which were produced by the code as written, but it does affect how the method should be described, and any reimplementing should fix the branch deliberately in one direction or the other.

The second measurement asks what the degree sequence actually encodes, since that is where the entire effect lives. Regressing the degree sequence of the base weights on the row norms and partitioning the variance, the answer differs by branch, and it differs in exactly the way the algebra predicts. Where $A = \sigma(|W|)$ and the entries are small, the logistic is close to linear, so the degree of a node is essentially the magnitude of its row and column; empirically the row norms explain 96% of the degree variance in layers 8 to 31 of `q_proj`. Where $A = \sigma(|WW^\top|)$, the degree is the total absolute alignment of a row with all other rows; there the row norms explain only 29% on average across the same layers of `v_proj`, and the remaining 71% is alignment structure that magnitude does not capture. The regulariser is therefore applying two different criteria depending on module shape: magnitude equalisation on square modules and alignment equalisation on non-square ones. This is consistent with the row-norm control being close to but

not equal to Omega-S, and it suggests a specific ablation, which we ran.

Which channel carries the effect: an ablation by module type. We restricted the penalty to one module type at a time while leaving the adapter unchanged on both, so that what varies is which channel is regularised and not where the model can learn. The comparison that matters is between the two ablated arms rather than against the full objective: both concentrate the same calibrated force on half as many modules, so they carry the same bias, whereas comparing either to the full arm would confound channel with intensity.

Seed	Only v (alignment)	Only q (magnitude)	Full objective
42	0.529	0.233	0.623
123	0.775	0.385	0.878
456	0.957	0.406	0.911
789	0.619	1.000	1.056
1011	1.000	0.296	0.708
2022	0.633	0.366	0.825
3033	0.800	0.342	0.837
4044	0.723	0.645	0.855
5055	0.621	0.800	0.741
6066	0.837	0.515	0.976
Mean	0.750	0.499	0.841
he_A (task-A capability)	0.289	0.193	0.287

Table 3: Retention with the penalty restricted to one module type, corrected orientation, ten seeds. The adapter is unchanged in both arms; only the set of modules the penalty acts on differs. **The last row is the reason this comparison must be read with care:** the magnitude arm ends the first task with 0.193 HumanEval against 0.289 for the alignment arm, so it learns task A substantially worse and its retention ratio is measured against a lower denominator. The direction of the difference is the same on the absolute measure, but the magnitude of the gap is not interpretable.

Two things follow, and they point in different directions. *The alignment channel contributes more:* it leads on nine of ten seeds (exact sign test, one-sided $p = 0.011$; Wilcoxon $W = 9$, two-sided $p = 0.065$, which does not clear the threshold because the single loss is also the largest difference), with a mean paired advantage of 0.116. *But neither channel reproduces the full objective:* the alignment arm exceeds it on three seeds of ten and the magnitude arm on two, and the better of the two (0.750) exceeds the no-regularisation baseline (0.631). The effect is not located in one channel; it requires acting on both module types.

We also note an observation we cannot yet interpret. The single seed on which the magnitude channel wins, 456, is the one where it wins by the largest margin of any seed in the table, and where it *exceeds the full objective* (1.148 against 1.023). On seed 5055 the two channels tie and both exceed the full objective. If this pattern held with more seeds it would suggest that different initialisations fall into different regimes, one magnitude-driven and one alignment-driven, which would be a candidate explanation for the bimodal per-seed advantage of Table 11 that our earlier structural measurement failed to find. On one seed of ten it is an observation and not a finding, and we report it as such.

One caveat bounds the whole ablation. Because the calibration fixes the total penalty gradient, restricting to half the modules concentrates that force on half as many, so each ablated arm receives roughly twice the accumulated force per module. This does not affect the comparison between the two ablated arms, which carry the bias equally, but it does affect any comparison against the full objective, and separating channel from intensity there would require a further run at reduced target.

3.4. How much run-to-run variation this setting carries

Every comparison in this paper is paired by seed, which presupposes that the seed identifies the run. We tested that presupposition by repeating an identical configuration: same seed, same code, same hardware, same arm. Four repetitions of one arm and three of another gave standard deviations of 0.068 and 0.104 in retention ratio, with individual pairs as far apart as 0.596 and 0.793. The same held when the four baseline arms were re-measured months after their first run. Three of the four reproduced their means to within 0.04; the per-seed values did not, with a standard deviation of 0.104 on the differences. The fourth is worth naming because it changed a comparison: EWC, whose mean rose from 0.593 to 0.691 on re-measurement. That is a shift of 0.098, the same order as the repeat-to-repeat deviation above, and it moved Omega-S from winning 8 of 10 seeds against EWC on the retention ratio to winning 6. Nothing about either method changed; the earlier EWC measurement simply sat low in its own distribution. This is the concrete reason every arm in Table 1 was measured in a single session, and it is what a comparison against a previously published baseline figure risks.

Two consequences follow and we state both. First, the presupposition does not hold in this setting: the seed fixes initialisation and data order but not the non-deterministic reductions in the GPU kernels, so a paired test across seeds is comparing runs that differ by more than the seed. Second, any mean difference below roughly 0.066, twice the standard error of a ten-seed mean, is not distinguishable here. That bound is what places the two orientations of M in the not-distinguishable category while leaving the comparisons against the baselines well outside it.

That GPU non-determinism dominates seed-to-seed variance is established elsewhere. Morin and Willetts [25] trained ResNet-50 fifty times with a fixed seed and attribute 74 to 87% of the observed standard deviation to it; Nagarajan et al. [26] report 12.4% relative deviation from the same source in reinforcement learning, and Pinto et al. [27] find significant weight and prediction differences between identically configured convolutional runs. We have not found the quantity reported for low-rank fine-tuning of language models, which is the regime here, and we report it because it bounds the reading of every seed-paired comparison in this literature, ours included. The robust design would average several repetitions per cell before pairing; we did not do that, and the cost of doing it is a multiplication of the compute budget by the number of repetitions.

3.5. One thing we would like others to run

This paper closes with several things untested, listed in Section 3.6. Most need GPU hours and we do not expect anyone to spend them on our behalf. One does not.

The mechanism measurement of Section 3.3 runs on base weights and takes minutes. It reports the elasticity of each of the four factors with respect to the weights, and it is what establishes that three of them are inert under the construction used here:

```
python experiments/check_M.py
```

We have run it on one model. The claim that the composite objective reduces in practice to a penalty on degree variance is a claim about *that construction on those weights*, and whether it holds elsewhere is the single thing that would most change what this paper can say. **A report that the factors are live on some other model is as useful to us as one that confirms they are not:** it would mean the reduction is specific rather than general, and we would rather know.

There is a reporting template in the repository.

3.6. Limitations

The primary limitation is scale: controlled structural experiments (degree variance, FLOPs reduction) were conducted on a small MLP with Split-MNIST. The main sequential fine-tuning result (mean retention 61.9% $\rightarrow$ 84.1% over tuned weight decay, ten seeds, 10/10 positive; 8/10 over tuned EWC) is measured on Llama-3-8B with LoRA adapters on public datasets using HumanEval as the retention metric, with the penalty verified connected to the autograd graph. Three caveats bound the interpretation. First, the per-seed advantage is bimodal (large wins, or losses only on the two seeds where EWC itself peaks); ten seeds establish the direction but not a per-seed structural model. Second, the mechanism is now measured (Section 3.3): the effect is carried by degree-variance reduction, not clustering, so “topological” describes the objective, not the operative mechanism in this formulation; how much of the degree-variance effect is separable from row-norm equalisation is now measured rather than assumed. A control swept on its own grid does not reliably beat no regularisation (6/10) and Omega-S exceeds it on 8/10, one-sided $p = 0.055$, which does not clear conventional significance at ten seeds. The control’s strength was moreover selected on two seeds that proved to be its own worst two, so another strength might serve it better. On GPT-2 (124M, full fine-tuning), a connected re-run finds that Omega-S combined with group lasso does not exceed group lasso alone in structured sparsity (-0.72% vs. -0.77% FLOPs). We report it because it bounds what the method should be claimed to do: whatever it contributes, it is not compression. The FSDP environment used PCIe rather than NVLink; overhead ratios are valid but should be re-confirmed in a higher-bandwidth environment.

A further limitation concerns what the name “Omega-S” covers. The retention results use the log-ratio composite and the pruning results use the raw trace penalty (Section 2.1); these are different objectives on different architectures and tasks, and the raw form is the arm that performs worst on retention. Neither result is invalidated by this, since each was produced by the objective it names, but the two should not be read as evidence for a single method until the composite is evaluated on the pruning setting as well.

Two further limitations follow from the work reported in Section B.12. The reference implementation builds the pseudo-adjacency as $\sigma(|WW^\top|)$ only for non-square W and as $\sigma(|W|)$ for square W , so the runs reported here mix two constructions, one per module type, while the Method section describes only the first. We report this because it changes how the method should be described and because the degree-variance term differs by roughly a factor of 400 between the two branches on `q_proj`; the numbers themselves are unaffected, having been produced by the code as written. And the negative result on the cosine reformulation is specific to catastrophic forgetting under LoRA on this model and task pair. It does not transfer to the index used as a monitoring statistic in domains where the graph is built from correlations rather than from weights, since there the clustering channel is not saturated to begin with and the index is not used to intervene on the network.

4. Conclusion

We have introduced Omega-S, a graph-based regulariser for large-scale neural networks whose objective is built, via Hutchinson trace estimation, from the clustering coefficient of the weight adjacency matrix $WW^\top$. The method is computationally efficient ($\mathcal{O}(N^2)$ per layer), adds negligible overhead to training (+1.5–3.7% at $K=10$), and requires zero inter-GPU communication in LoRA+FSDP settings. In sequential fine-tuning of Llama-3-8B, the log-ratio form of Omega-S leaves more code capability than no regularisation at all on 9 of 10 seeds (0.173 $\rightarrow$ 0.238 absolute HumanEval, +37.7% relative, $p = 0.011$), and leads weight decay on 9 of 10 and EWC on 8 of 10 in retention (61.9% $\rightarrow$ 84.1%), with zero access to previous-domain data. A first five-seed run showed the same direction more weakly, so the effect strengthened with sample size.

We are explicit about mechanism. Although the objective is topological by construction, a direct measurement shows that in this formulation the penalty acts through reduction of node-degree variance, not clustering: the clustering channel is saturated by the logistic map used to build the adjacency and does not move. Omega-S as it stands is thus best understood as a degree-variance regulariser with a (currently inert) topological objective; whether its benefit is separable from row-norm equalisation is now tested rather than deferred, and a control swept on its own grid does not reproduce the gain, though the margin over it does not clear conventional significance at ten seeds. This revises the explanation of the method, not its measured utility, which exceeds both standard baselines. Restoring the topological channel through a contrast-preserving construction was the obvious repair and we built it: it does revive the clustering channel and it makes retention worse on all ten seeds (Appendix A). Whether some other construction would do better is open, but the one that seemed most promising a priori does not.

A further limit concerns the scope of the regulariser. Weight decay, used here as a retention baseline, is not primarily a forgetting method: it is the standard tool for generalisation on a single task. Whether Omega-S also competes with weight decay on that ground, that is, whether degree-variance control improves single-task generalisation and not only sequential retention, is a coherent follow-up we have not tested and flag without prejudging the outcome.

Finally, Omega-S reflects a broader principle: structural regularisation principles observed in biological systems can provide productive priors for artificial neural network design. The formalisation of this principle, which runs from soil microbiome network observations [1, 2] to a mathematically grounded and empirically validated regulariser for LLMs, represents the central intellectual contribution of this work.

Patent. The Omega-S method is the subject of a pending patent application.

Code Availability and Licensing

The complete PyTorch implementation of Omega-S is publicly available at <https://github.com/BiomeMakers/OmegaS-LLM>.

Dual Licensing: The source code is released under the **GNU Affero General Public License v3 (AGPL-3.0)** for non-commercial academic research, education, and non-profit experimentation. AGPL-3.0 permits free use, modification, and redistribution with attribution, provided that any network deployment of a modified version includes public disclosure of the modified source code.

Commercial licensing, including production deployment, integration into commercial training pipelines or any use within a for-profit organisation, requires a separate commercial licence that includes both software and patent rights. Inquiries regarding commercial licensing, research partnerships, or collaborative development agreements should be directed to the author via the repository.

Appendices

The material below is not needed to use the method or to judge the results. It contains the framing that motivated the work, the related-work survey, the experiments that support secondary claims, and the full record of what we tried and it did not work. We keep it because the negative results are part of what the paper reports, not because a reader needs them first.

Note on the appendices

The main text reports the configuration this paper uses: the log-ratio composite with $M = 1/\lambda_2$, with every arm measured in a single session. Several appendices predate that configuration and report the alternative orientation $M = \lambda_2$. They are: the calibration sweep that selected the penalty strength; the *strength sweep* of the row-norm control, which is a separate experiment from the row-norm arm of Table 1 and asks what that control achieves when tuned on its own grid; the hyperparameter selection protocol for weight decay and EWC; and the cosine construction study. We have not re-run them because what each establishes does not depend on the orientation: the calibration sweep locates a target that both configurations use, the row-norm sweep tests whether the effect reduces to norm balancing, and the cosine study asks whether a construction that revives the clustering channel helps, which Section 3.3 shows it does not under either orientation.

The arms of Table 1 are not among them. No regulariser, weight decay, EWC, the row-norm control and the cosine composite were all re-measured in one session with the configuration this paper uses, using the hyperparameters those earlier sweeps had selected. What was not repeated is the *selection* of those hyperparameters, not their evaluation.

Figures that differ between the two configurations are dated by which appendix they appear in. Where an appendix reports a mean retention of 76.6% for Omega-S, that is the alternative orientation; the configuration of this paper gives 84.1%. We say this once here rather than repeating it in each appendix.

A. The row-norm control, swept on its own grid

This appendix reports the alternative orientation $M = \lambda_2$; see the note above. The row-norm arm of Table 1 is separate and was measured with the configuration of this paper.

Not equivalent to norm-balancing: a control, and a protocol defect. A recurring concern for any penalty on $WW^\top$ is that it may reduce, in practice, to equalising the norms of the weight rows, since degree and row norm are strongly correlated. This is the objection that decides how novel the mechanism is, so we tested it rather than argued about it.

Our first attempt at the test failed, and the failure was ours. The tuning phase of the ten-seed run swept weight decay, EWC and the two Omega-S arms; it did not sweep the row-norm control. In the evaluation phase that control was launched with the calibration target selected for Omega-S, a value chosen to make a *different* penalty a fixed fraction of the task gradient. The control was therefore the one arm never given a strength of its own, and it collapsed (retention 20.7%, absolute HumanEval 0.027, against 0.153 already after the first task, so the model was damaged rather than merely forgetful). Beating it established nothing. We found this by reading the orchestration script rather than by reasoning about the numbers, and we report it because an earlier version of this paper drew a conclusion from that arm which the arm could not support.

We then ran the control properly, on a grid of its own, using the same two selection seeds and the same ten evaluation seeds as every other arm. The grid was placed below the Omega-S grid, since 0.03 was already known to destroy this penalty (Table 4). The selected target was 0.001, at the lower edge, which for this arm is the degenerate limit rather than an unexplored optimum: as the target approaches zero the penalty vanishes and the arm converges to no regularisation by construction.

The result is a separation, and a qualified one. Tuned on its own grid, the control does not reliably exceed no regularisation at all (6 of 10 seeds, mean absolute 0.174 against 0.168, one-sided sign test $p = 0.38$), and it carries the highest variance of any arm in the study (retention standard

Table 4: Row-norm control, own hyperparameter sweep, on the same two selection seeds (42, 123) used for every other arm. Retention rises monotonically as the penalty weakens, which for this arm is convergence toward no regularisation rather than an interior optimum.

Calibration target	Retention	Std
0.001	51.9%	0.065
0.003	39.6%	0.192
0.01	31.5%	0.091
0.03	28.3%	0.223

Table 5: Per-seed absolute code capability (HumanEval pass@1 after the prose task) for no regularisation, the row-norm control at its own selected strength, and Omega-S. The last column is Omega-S minus the control.

Seed	None	Row-norm	Omega-S	Δ
42	0.134	0.146	0.152	+0.006
123	0.226	0.165	0.195	+0.030
456	0.165	0.171	0.274	+0.103
789	0.177	0.244	0.207	-0.037
1011	0.134	0.195	0.177	-0.018
2022	0.201	0.183	0.299	+0.116
3033	0.177	0.159	0.256	+0.097
4044	0.152	0.195	0.238	+0.043
5055	0.171	0.098	0.207	+0.109
6066	0.146	0.189	0.220	+0.031
Mean	0.168	0.174	0.223	+0.048

deviation 0.216 against 0.145 for Omega-S, with one seed collapsing to 0.254). Omega-S exceeds it on 8 of 10 seeds, one-sided $p = 0.055$. Row-norm equalisation on its own therefore does not buy retention in this setting, whereas Omega-S does.

Two limits bound that reading and we state both. First, at ten seeds the margin of Omega-S over the control does not clear conventional significance; it sits at the same level as the margin over EWC. Second, the control’s strength was selected on seeds 42 and 123, which turned out to be its own two worst of the ten (0.453 and 0.509 retention, against 0.583 to 0.914 elsewhere), so its selection is unreliable and another strength might serve it better. We did not re-select on the evaluation seeds, because that would hand the control a best-of-several advantage on the evaluation set which Omega-S never had.

The effective form is decisive. The raw $\text{Tr}((WW^\top)^3)$ penalty applied directly to LoRA factors (the form used in an earlier iteration of these experiments) is the weakest arm in the same ten-seed sweep (53.9% mean retention, below the 63.1% no-regularisation baseline, winning only 1/10 against weight decay). The log-ratio **StochasticOmegaS** form is what delivers the results above. We attribute the gap to scale sensitivity of the raw sextic penalty, whose gradient vanishes as weights shrink; the log-ratio form is approximately scale-invariant. Consequently, the library and the experiment scripts are being unified on the log-ratio form, and earlier internal figures that relied on the raw form are superseded.

B. Background and Related Work

B.1. Topological Properties of Weight Graphs

Neural network weight matrices can be interpreted as bipartite graphs where neurons are nodes and weights define edge strengths. When the degree distribution is highly skewed, with a small number of neurons capturing most of the total norm, we term these high-degree nodes *weight monopolies*.

The clustering coefficient, proportional to $\text{Tr}(A^3)$, quantifies the density of local connectivity clusters. We use it as a structural descriptor and do not read a high value, on its own, as the signature of a weight monopoly; as noted above, it is the degree variance that distinguishes distributed from concentrated triadic structure.

B.2. Scale-Free Networks in Biological Systems

The scale-free network model [12] describes graphs where the degree distribution follows a power law $P(k) \sim k^{-\gamma}$. Such networks exhibit high resilience to random node failures while remaining vulnerable to targeted attacks on hubs.

Studies by Biome Makers and collaborators have characterised the topological structure of microbial interaction networks across diverse agricultural conditions. Ortiz-Álvarez et al. [1] demonstrated that farming practices in vineyard soils produce measurable disturbances in the network properties of local fungal communities, with structural monopolies emerging under high anthropogenic pressure. At global scale, Saati-Santamaría et al. [2] showed that soil microbial diversity and network organisation respond consistently to land use and agricultural inputs worldwide, establishing degree distribution and hub structure as reliable indicators of ecosystem health across biomes. These topological properties are further formalised as diagnostic tools in two Biome Makers patents: one on evaluating ecological disturbance via network properties of organism communities [4], and one that defines resilience indices derived from the *transitivity* of bacterial and fungal co-occurrence networks [3]. The latter is the direct antecedent of the present work. One qualification is needed for accuracy: the *global* transitivity of a graph is proportional to $\text{Tr}(A^3)$, whereas the vineyard study computes the average of the *local* transivities, which captures the same triadic closure but is a different functional and is not proportional to $\text{Tr}(A^3)$. The lineage runs through the notion of triadic closure, not through an identity of statistics.

B.3. Hutchinson Trace Estimator

The Hutchinson estimator provides an unbiased stochastic approximation of the trace of a matrix M :

$$\text{Tr}(M) \approx \frac{1}{n} \sum_{i=1}^n z_i^\top M z_i, \quad z_i \sim \text{Rademacher}(\pm 1). \quad (6)$$

Applied to $M = A^3 = (WW^\top)^3$, the estimator avoids materialising A by decomposing the matrix-vector product:

$$A^3 z = W \left(W^\top \left(W \left(W^\top \left(W(W^\top z) \right) \right) \right) \right). \quad (7)$$

Each step requires $\mathcal{O}(mn)$ operations, yielding total cost $\mathcal{O}(6n_p mn)$ per layer versus $\mathcal{O}(m^3)$ for explicit A^3 . The Hutch++ variant [13] provides variance reduction for ill-conditioned matrices.

B.4. Related Regularisation Methods

Weight Decay / L2 pushes all weights uniformly toward zero, agnostic to structural role. As we show empirically, it produces unstructured sparsity not amenable to FLOPs reduction via structural pruning.

SAM [5] seeks flat minima in the loss landscape and is blind to the internal topology of weight matrices. Omega-S and SAM are complementary, not competing.

Graph-spectral regularisers. Fiedler regularisation [6] penalises the algebraic connectivity λ_2 of the graph induced by $|W|$ during training, with theoretical support from spectral graph theory and an equivalent formulation as a structurally weighted L_1 penalty. It is the closest antecedent to the present work in kind: a functional of a graph built from the weights, minimised as part of the training objective rather than computed as a diagnostic. Two things separate the two. First, the objective here is triadic, built on $\text{Tr}(A^3)$, and λ_2 appears only as one of four factors; a graph-theoretic penalty defined on a bipartite construction, as several in this family are, would annihilate the triadic term identically. Second, and against our own interest, our measurement (Section 3.3) finds the λ_2 channel numerically inert in this construction, so we make no claim to improve on Fiedler regularisation on its own ground; the comparison remains to be run.

Structured / Group Sparsity [11] explicitly targets neuron elimination based on weight magnitude. In our own measurements the composite penalty combined with group lasso does not exceed group lasso alone in the compression achieved, which is why we make no claim in that direction here.

B.5. Geometric and Spectral Approaches to Continual Learning

A parallel and rapidly developing line of work attacks catastrophic forgetting by reasoning about the *geometry* of weight matrices rather than the magnitude of individual parameters. Muon-OGD [22] argues that Frobenius-norm geometry is not the natural choice for matrix-valued LLM parameters, and reformulates projection-based continual learning under spectral-norm geometry, reporting gains on the standard and 15-task continual-learning benchmarks and on TRACE [36]. Related efforts constrain the spectral norm and stable rank of routing manifolds in mixture-of-experts architectures [23], or penalise functional drift within the top- k eigenspace of a task kernel [24]. Our premise, that the topology of a weight matrix and not merely the size of its entries governs how new learning overwrites old, is shared by this literature and independently supported by it.

A geometric law for forgetting, and a measurement of ours that speaks to it. Steele [28] gives a geometric account of forgetting in LoRA specifically, reporting that it follows $\mathcal{F} = \alpha(1 - \cos^2 \theta_{\min}) + \beta$, where $\theta_{\min}$ is the minimum principal angle between the task gradient subspaces, validated on synthetic tasks, Split-CIFAR100 with ViT-LoRA and sequential GLUE with RoBERTa-LoRA. Two consequences of that law bear on the present work.

The first is that we can report a measurement of the governing quantity on a model of the scale evaluated here. In a separate screen we computed the principal angles between the row spaces of two rank-8 LoRA deltas trained on different tasks from a common Llama-3-8B backbone, across all sixty-four attention projections. Against a simulated null for two random eight-dimensional subspaces in $\mathbb{R}^{4096}$ (leading cosine 0.077 on average, 0.100 at the maximum over three hundred draws), the leading principal cosine is 3 to 5 times the null in fifty of the sixty-four modules, while the second and all subsequent angles are indistinguishable from it. The overlap between two adapters trained on different tasks is therefore essentially rank one: they share one direction, and the rest of their agreement is chance. It also varies systematically with depth, being largest in the first and last layers and smallest in the middle, and five modules sit exactly at the null.

The second is a positioning point. Steele reports that orthogonalisation methods such as O-LoRA add little when natural orthogonality is already high. In our measurement seven of the eight directions are at the null, that is already near-orthogonal, so on that reading an orthogonalisation approach would have limited room in this setting. Omega-S does not orthogonalise; it acts on the degree distribution of the weight graph, which is a different quantity, and this is one reason

the two families are complementary rather than competing.

An objection from function space, which we record rather than answer. Hidekel and Raviv [24] argue in the NTK regime that forgetting concentrates in a small number of old-task kernel eigenmodes, and that this explains why regulariser acting in *parameter* space can miss interference in *output* space. Omega-S is a parameter-space regulariser, so the objection applies to it directly and we state it rather than passing over it. Two things bound it in our setting without dissolving it. Their claim is that parameter-space methods can miss such interference, not that they cannot help: the retention results reported here are obtained by a parameter-space penalty and hold on nine of ten seeds. And their account predicts where a ceiling should lie, which is a testable statement about our method that we have not tested. Measuring whether the residual forgetting under Omega-S concentrates in the old-task kernel eigenmodes their theory identifies would establish whether the two accounts describe the same limit, and we identify it as an experiment we have not run.

A sharper version of the same objection, and what separates our method from its target. Ning et al. [29] make the empirical case that weight-space regularisation underperforms on large language models because the models are polysemantic: per-weight importance estimates of the kind EWC computes are too coarse to isolate the knowledge that needs protecting. They report that task-relevant representations are linearly separable in a sparse-autoencoder feature basis but indistinguishable from chance in the weight basis, and that weight-space protection is close to non-selective at the concept level, and they regularise in activation space instead. This is the most direct challenge to the family Omega-S belongs to that we are aware of, and we would rather engage it than omit it.

The engagement is a distinction, not a rebuttal. Their evidence concerns *per-weight importance*: the claim is that no scalar attached to an individual weight identifies what a concept occupies. Omega-S attaches no importance to individual weights. It penalises a scalar property of the *whole matrix*, the variance of the degree sequence of the graph it induces, which is a statement about how connectivity is distributed rather than about which weights matter. Whether the polysemanticity argument extends from per-weight importance to distributional properties of the weight matrix is, as far as we can tell, open, and it is a question their framework is better placed to settle than ours. We note only that the retention results reported here were obtained in the weight basis, which the argument as stated would not predict.

A comparison of magnitude worth recording. Independently, Laitinen-Fredriksson Lundstrom-Imanov [30] report a mechanistic study of forgetting across twenty models and find that a curvature-regularisation intervention reduces forgetting by 34% with minimal impact on new-task learning, a figure close to the 33% relative gain in retained capability reported here. The two are not commensurable, the settings, models and metrics differing throughout, and we place them side by side only to indicate that interventions acting on the geometry of the loss or of the weights appear to reach effects of a similar order, which is information a reader deciding whether this line is worth pursuing should have.

Two method classes, and where Omega-S sits. It is useful to separate two families. *Penalty* methods add a term to the loss that discourages changes deemed harmful: EWC [31] weights those changes by Fisher information, LwF [32] by output distillation. *Projection* methods instead modify the update direction itself, restricting it to a subspace orthogonal to protected past-task directions: OGD [33], O-LoRA [34], OSFT and Sculpting Subspaces [35], and Muon-OGD. On the multi-task benchmarks reported in [22], projection methods substantially outperform penalty methods: 75.8 (O-LoRA) and 78.9 (Muon-OGD) against 50.3 (EWC) and 52.3 (LwF) average accuracy on the standard benchmark. Omega-S belongs to the penalty family, and we state this explicitly rather than leaving it to be inferred.

Scope of the comparison we do and do not make. We do not report a head-to-head comparison against these methods, and we are precise about why. The setting evaluated here is single-adapter LoRA fine-tuning with retention measured on a held-out capability after a task pair (Section 3.2), whereas the works above evaluate multi-task streams of five to fifteen tasks under full or multi-adapter fine-tuning. The numbers are therefore not commensurable, and quoting them side by side would be misleading. What follows from this is an obligation rather than a defence: extending Omega-S to those benchmarks, with EWC as the class-appropriate baseline and O-LoRA or OSFT as the strong incumbents, is the natural next validation step, and it is specified as *Experiment A* in the distributed validation protocol accompanying this work.

An open problem where Omega-S may be directly relevant. Projection methods must decide *which* directions to protect. Current practice, including Muon-OGD’s low-rank instantiation, takes the top- k left and right singular vectors of the pretrained weight matrix, and [22] names identifying these constraint directions more effectively as explicit future work. The per-node decomposition $\text{diag}(A^3)$ used by Omega-S ranks units by a different criterion: the concentration of closed three-step walks incident on each unit, which is a function of the eigenvectors as well as the eigenvalues, and is therefore not recoverable from the singular-value spectrum alone. In preliminary probing on TinyLlama-1.1B we find this ranking to be largely uncorrelated with Fisher-information importance (mean Spearman $\rho \approx 0.02$ across 88 attention projections, after controlling for row norm), indicating that it carries information distinct from the standard parameter-importance signal. Whether it also carries information distinct from the dominant singular directions, and whether substituting or augmenting the protected subspace with it improves retention, is untested. We flag it as a concrete point of contact with this literature and an open invitation to collaboration rather than as a result.

B.6. Non-Uniform Structured Pruning of LLMs

A growing body of work demonstrates that uniform sparsity across transformer layers is suboptimal [14], motivating *non-uniform* sparsity allocation strategies that assign differentiated pruning budgets per layer.

OWL [14] allocates higher sparsity to layers with fewer activation outliers, mapping the outlier distribution linearly to sparsity ratios. It demonstrates consistent improvements over uniform Wanda and SparseGPT at high sparsity levels.

AlphaPruning [37] estimates per-layer importance using Random Matrix Theory, specifically the heavy-tailed spectral exponent of the weight matrix. Layers with heavier-tailed spectra are considered more important and receive lower sparsity. This is the closest prior work to Omega-S in spirit: both use a structural property of the weight matrix (rather than activation statistics) to guide pruning.

SV-NUP [38] quantifies per-layer contribution via Shapley values, enabling tailored pruning budgets that preserve the most critical layers. Notably, [38] also cites topological insights into restricted Boltzmann machines [40] as motivation, the same class of structural observations that underlies Omega-S.

DarwinLM [39] uses evolutionary search to discover optimal non-uniform sparsity allocations, trading off computation for accuracy.

Distinction from Omega-S. These methods share with Omega-S the premise that weight-matrix structure carries actionable information, but they differ in *stage*, *objective*, and *quantity*. OWL and AlphaPruning operate *after* training as allocators of a compression budget; their target is inference efficiency at fixed accuracy. Omega-S instead acts *during* training as a regulariser on the topology of each weight matrix, and its target is *knowledge retention*, that is, resistance

to catastrophic forgetting under sequential fine-tuning, rather than compression. Where AlphaPruning *measures* the heavy-tailedness of a layer’s spectrum to decide how much to prune, Omega-S *reshapes* the weight-graph structure during optimisation to reduce its concentration (degree variance). The two directions are therefore complementary rather than competing: a model regularised with Omega-S can still be pruned afterward by OWL or AlphaPruning, and the questions they answer, namely *how much of each layer to remove* versus *how to train weights that forget less*, do not overlap. We do not claim Omega-S as a pruning-allocation signal; its contribution is at training time, on retention.

B.7. Conceptual Origin: From Soil Microbiomes to Neural Networks

The conceptual origin of Omega-S lies not in the loss landscape geometry of deep learning, but in the observed topological properties of biological information networks, specifically the connectivity structure of microbial communities in agricultural soil ecosystems.

Empirical studies of soil microbiome networks report that agricultural disturbance systematically reorganises the association structure of the community. In a survey of 350 vineyard soils across two countries [1], soils under low-intervention management showed a higher clustering coefficient, lower modularity and a lower proportion of co-exclusion edges than conventionally managed soils, a combination the authors read as denser, less niche-partitioned communities with greater resistance to species loss. In a global survey of 1,921 soils from 33 countries [2], agricultural inputs consistently reduced network density across both bacterial and fungal communities, which the authors associate with reduced functional redundancy.

We are deliberate about what we do and do not take from this. The two studies use different association estimators, different units of analysis (one network per sample against one network per condition) and different clustering definitions, so their individual factors are not directly comparable and we do not pool them into a single directional claim. Neither reports degree distributions or power-law fits, so we make no scale-free claim here; an earlier version of this paper did, and it was not supported by the sources cited. What the two share is that disturbance reorganises network structure in directions their authors link to lower resilience. The intuition we carry across is accordingly a weak one: how connectivity is *distributed* over a network, and not only how much of it there is, carries information about robustness that node counts and diversity indices do not.

This biological observation raises a compelling question for artificial neural networks: do similar structural monopolies emerge during training, and if so, do they carry analogous costs in terms of robustness, generalisation, and efficiency?

The answer, we argue, is yes. During the training of large language models, a small subset of neurons progressively concentrates disproportionate weight connectivity across layers, creating highly non-uniform degree distributions in the implicit weight graph. This concentration, which we term *weight monopolies*, persists even under strong weight decay and sharpness-aware optimisation, which target the loss landscape geometry but leave the internal topology of weight matrices untouched.

B.8. Thermodynamic and Information-Theoretic Framing

The connection between biological network topology and artificial neural networks has a deeper thermodynamic grounding. In physical systems, the principle of minimum energy dissipation (related to Boltzmann’s entropy formulation) predicts that stable, efficient systems minimise redundant energy flows by distributing load across available channels. Structural monopolies represent thermodynamically inefficient configurations: they concentrate functional load on a subset of nodes while leaving the remaining capacity idle.

The clustering coefficient of a network, quantified as the normalised count of closed triangles in the connectivity graph and proportional to $\text{Tr}(A^3)$, is a sensitive measure of how triadic structure is distributed. We deliberately do not attach a direction to it. The same triangle count is compatible with connectivity spread evenly across the graph and with connectivity concentrated on a hub and its core, and what separates those two configurations is the variance of the node degrees rather than the triangle count itself (Section 3.3 and [21]). What we retain here is only that $\text{Tr}(A^3)$ is a structural quantity distinct from parameter magnitude, and that its gradient is cheap enough to apply during training.

This framing connects Omega-S to a broader principle: just as the curvature of spacetime governs the trajectories of physical systems in general relativity, the topological curvature of the weight connectivity graph governs the flow of information through a neural network. Standard regularisers operate on the metric of the loss landscape; Omega-S operates on the metric of the network’s internal topology.

B.9. Relation to Spectral Quantities

We state a fact about $\text{Tr}(A^3)$ that a reader may otherwise raise as an objection, and explain why it does not undermine the use made of it here. For symmetric A with eigenvalues $\{\lambda_i\}$,

$$\text{Tr}(A^3) = \sum_i \lambda_i^3, \quad (8)$$

so the quantity is exactly the third moment of the spectrum of $WW^\top$. It is therefore not an alternative to spectral analysis but a particular functional of the spectrum, in the same family as the effective rank, the Vendi score and the Absorption Ratio. As a *descriptive statistic* it carries no information those measures lack, and we make no claim that it does; the companion theoretical work develops this point and its consequences [21].

The relevant observation is that Omega-S is not used here as a statistic. It is used as a *training objective*, and what an objective contributes is its gradient, not its value. Different functionals of the same spectrum have different gradients and therefore induce different geometries:

$$\nabla_W \text{Tr}\left((WW^\top)^3\right) = 6(WW^\top)^2 W, \quad (9)$$

a polynomial expression evaluable through matrix–vector products alone. The entropy-based functionals require the eigenvalues themselves, so differentiating them entails an eigendecomposition at every optimisation step, which is expensive at LLM scale and numerically delicate where eigenvalues are near-degenerate. Combined with the Hutchinson estimator, which avoids materialising A at all, this is what makes the penalty tractable as a per-step term rather than a diagnostic computed occasionally. The contribution claimed here is accordingly narrow and specific: not that penalising spectral structure is novel, but that *this* functional, with *this* estimator, is cheap enough to apply during training and improves retention when so applied.

A second consequence of Eq. (8) concerns where the non-redundant information in the framework resides. The global scalar is spectrally determined; the per-node decomposition $\text{diag}(A^3)$ is not, since $(A^3)_{ii} = \sum_k \lambda_k^3 (v_k)_i^2$ depends on the eigenvectors. This is the basis of the attribution discussed in Section B.5.

B.10. Structural Control: Omega-S vs. Weight Decay

Omega-S achieves $8\times$ greater reduction in degree variance compared to weight decay (0.136 vs. 1.12), with only 0.54 pp accuracy cost versus 2.90 pp for weight decay. Weight decay reduces the maximum hub more aggressively (69.00 vs. 116.83), consistent with indiscriminate norm suppression. We interpret this as evidence that the two methods target different aspects of weight

Table 6: Structural control metrics across training configurations.

Architecture	Eval Loss ($\downarrow$)	Accuracy ($\uparrow$)	Degree Var. ($\downarrow$)	Max Hub ($\downarrow$)
Baseline	0.1650	95.46%	41.94	125.47
Omega-S	0.1894	94.92%	0.136	116.83
Weight Decay ($\lambda=0.01$)	0.2464	92.56%	1.12	69.00

concentration: Omega-S redistributes the global degree distribution while preserving hubs that serve legitimate routing functions, the neural analogue of the microbial network observation that not all hubs are monopolies.

B.11. How the hyperparameters were selected, and the anomaly this explains

Table 1 contains a result a reader will notice, and we explain it rather than leave it standing: both regularised baselines retain *less* than the arm with no regulariser at all (59.3% and 59.3% against 63.1%). The explanation is the selection protocol, which is weaker than an earlier version of this paper described.

Every arm was selected, and we give the grids. Weight decay was swept over $\{0, 0.01, 0.05, 0.1\}$; the grid included zero, so the procedure was free to switch the regulariser off, and it did not, returning $\lambda = 0.05$. EWC’s λ was swept over $\{10^2, 5 \times 10^2, 10^3, 5 \times 10^3, 10^4\}$ and returned 10^3 , which was both the highest-scoring value and by some margin the most stable across the selection seeds (standard deviation 0.011, against 0.06 to 0.08 for the others). That value coincides with the library default, so the sweep confirmed the default rather than changing it; a concern recorded at the time, that EWC might be running at a badly set λ , was checked and dismissed rather than left standing. Omega-S’s calibration target was swept over $\{0.03, 0.1, 0.3, 0.5\}$ and returned 0.03.

The limitation is therefore not that the arms went unselected. It is that the selection was made on *two* seeds, and both of them are among the ten later used for evaluation, so the selection set is a subset of the evaluation set rather than disjoint from it. The retention observed during selection (0.69 for weight decay, 0.72 for EWC) is well above what the same arms attain over the full ten seeds (0.59 and 0.59), which is the regression one expects when a small and favourable selection set is expanded, and it is why both tuned arms can sit below the no-regulariser arm without either method being at fault.

Three consequences follow, and we keep them separate.

First, the two baselines in Table 1 should be read as one reasonable fixed setting of each method, not as the best each method can do. A properly selected weight decay would plausibly land at or above the no-regulariser baseline, and the +17.3 pp margin over it should not be quoted as though it were.

Second, the comparison that does not depend on this protocol is the one against the no-regulariser arm, which has no hyperparameter to select and is therefore unbiased on these ten seeds. Omega-S beats it on **8 of 10 seeds**, mean 63.1% $\rightarrow$ 76.6% (+13.5 pp; exact sign test, one-sided $p = 0.055$, two-sided $p = 0.11$). We regard this as the conservative headline and prefer it to the margin over weight decay. For completeness the same test gives 9/10 against weight decay (one-sided $p = 0.011$) and 8/10 against EWC (one-sided $p = 0.055$).

Third, Omega-S’s own selection carries the same weakness, and one of its own: it was selected on the same two seeds, and the value returned, 0.03, sits at the edge of the swept range, with retention falling monotonically as the target rises. A grid whose optimum lies on its boundary has not been shown to contain the optimum. Running the selection on seeds disjoint from the

evaluation set, and extending the Omega-S grid below its current floor, are the first two things we would change, and where a replication should differ from us.

B.12. Testing the contrast-preserving reformulation: a negative result

The mechanism measurement above left one question unresolved: the clustering channel is inert only because of the saturating σ , so a construction that preserves contrast would let a genuinely clustering-driven variant be tested, and whether such a variant outperforms the degree-variance mechanism is unknown. We subsequently built that variant and tested it. This section reports the outcome, which is negative and, we think, informative: activating the clustering channel does not improve retention, it degrades it, and it does so monotonically.

Which constructions we screened, and why this one. Restoring an informative clustering channel means replacing the bounded map that destroyed it, and several candidates are available. We screened six on the base weights before committing to any of them, using the excess statistic C/D , which equals exactly 1 for a constant affinity matrix and exceeds 1 in the presence of triadic structure. The screen is cheap, runs on a laptop, and falsifies most of the candidates outright.

Construction	Form	Excess C/D
Original (reference)	$\sigma(WW^\top)$	1.0000
Original, zero diagonal	as above, $A_{ii} = 0$	1.0000
Temperature-scaled	$\sigma(WW^\top /\tau - b)$	1.0000 for all $\tau \in [0.1, 50]$
Degree-normalised	$D^{-1/2}AD^{-1/2}$	0.9997 (spread 0.0035)
Rank-normalised	empirical CDF of $ WW^\top $	decompresses (C : 1.0 $\rightarrow$ 0.69)
Row-wise cosine	$ \cos(w_i, w_j) $, $A_{ii} = 0$	1.357 median, 28 modules
Doubly stochastic	Sinkhorn scaling of $ WW^\top $	1 by construction

Table 7: Screening of contrast-preserving constructions on base Llama-3-8B weights. Zeroing the diagonal does not help, which rules out diagonal dominance as the cause. Temperature scaling fails at every temperature because $|WW^\top|$ spans many orders of magnitude, so dividing by τ does not prevent saturation. Degree normalisation, the option with the strongest prior support in the clusterability literature, also fails: normalising a matrix the logistic has already flattened does not restore contrast, because the problem is one of scale upstream. Two candidates survive, and the cosine form was chosen over the rank form because the rank map is piecewise constant and therefore gives a noisy or vanishing gradient. The doubly stochastic form is listed for completeness: it nulls the excess exactly, which makes it useful as an attribution control rather than as a training construction.

Two further observations from the same screen bear on the choice. First, elasticity: under the original construction the median elasticity of $\text{Tr}(A^3)$ with respect to a perturbation of W is 0.84, and in at least one module it is exactly zero in float32, meaning the penalty cannot move the objective there at all. Under the cosine form the median is 16.6, roughly twenty times more responsive. Second, the signal is not uniform across module types: on `down_proj` the cosine excess is 1.3% at the median against 36% in attention, and it decays monotonically with depth (11.3% at layer 0, 0.33% at layer 31), which is why the experiments that follow apply the construction to the attention projections.

Construction. Of the six candidate constructions screened above, the one that both removes the saturation and admits a clean algebraic argument is the row-wise cosine affinity

$$A_{ij}^{\cos} = |\cos(w_i, w_j)| = \frac{|\langle w_i, w_j \rangle|}{\|w_i\| \|w_j\|}, \quad A_{ii}^{\cos} = 0. \quad (10)$$

Two properties motivate it. First, it is invariant to row scaling by construction: normalising a row changes no cosine, so the node degree k_i becomes mathematically independent of $\|w_i\|$. This is the property the original construction lacks, where k_i contains $\|w_i\|^2$ through the diagonal of $WW^\top$. Second, the entries occupy the full $[0, 1]$ range rather than collapsing to the midpoint of a bounded map.

Static screening. Before spending compute we verified on the base weights of Llama-3-8B that the construction actually decompresses the clustering channel. We use the excess statistic C/D , which equals 1 exactly for a constant affinity matrix and exceeds 1 in the presence of triadic structure. Under the original construction $C/D = 1.0000$ to four decimal places in every module examined, and in at least one module the elasticity of $\text{Tr}(A^3)$ with respect to a perturbation of W is exactly zero in float32: the channel is not merely weak, it is numerically dead. Under Eq. 10 the same statistic gives a median excess of 1.36 across 28 modules (per-module range 1.05 to 1.84), and the elasticity is roughly twenty times larger. Measured with the library’s own normalisation $\text{Tr}(A^3)/\|A\|_F^3$, which is the quantity that actually enters the objective, the clustering term moves from 0.9999 to 0.802 in `q_proj` and from 0.99999 to 0.512 in `v_proj`. The channel is live.

The channel is controllable under training. Static structure need not survive optimisation, so we screened that separately before evaluating retention. Training with a clustering-only penalty $\pm\lambda\log\text{Tr}(A_{\text{cos}}^3)$ and diagnosing the exact excess statistic every 25 steps, the sign of the penalty separates the direction of movement in every module and at every dose tested: a positive coefficient lowers C/D and a negative one raises it, in 16 of 16 measurements across `q_proj`, `k_proj`, `v_proj` and `o_proj`. The between-sign contrast grows monotonically with penalty strength. The clustering channel is therefore not only live in the weights, it is steerable during LoRA training, which the original construction is not.

Recalibration. The ratio calibration fixes the penalty gradient at a target fraction of the task gradient, so it normalises magnitude but not direction. Because the cosine construction changes the direction of the penalty gradient, the optimal target does not transfer. Swept on two seeds held out of the evaluation set, the optimum is interior and identical in rank order on both seeds ($0.003 > 0.01 > 0.03 > 0.001$), an order of magnitude weaker than the optimum of the original construction. We note that the original sweep selected its target at the edge of its grid; extending the grid downwards here is what allowed the optimum to be bracketed on both sides.

Retention outcome. We then evaluated the full composite objective with the cosine construction, changing nothing else: the same four factors, the same formulas, the same modules, the same ten evaluation seeds. Against the original formulation it loses on **all ten seeds** (mean retention 0.537 against 0.766; mean paired difference -0.229 ; Wilcoxon signed-rank $W = 0$, two-sided $p = 0.002$). Every individual value of the cosine arm falls below every individual value of the original arm, so the comparison does not depend on the pairing. Against weight decay and EWC the cosine arm is a statistical tie with the point estimate below (4/10 each, mean difference -0.056), and it also falls below the no-regularisation arm.

Decomposition. Running the clustering-only arm on the same ten seeds at the same operating point isolates the contribution of the remaining three factors. Their median contribution is $+0.147$ under the cosine construction against $+0.227$ under the original one, that is roughly two-thirds as much, while the clustering-only arm itself drops from 0.539 (original construction, where the term is inert) to 0.477 (cosine, where it is active). The arithmetic is consistent: the loss is accounted for by the clustering term changing from inert to actively harmful, plus a smaller reduction in what the other factors contribute. We note that the clustering-only arm has by

far the highest between-seed variance of any arm we have run (standard deviation 0.35, range 0.09 to 1.33), so the pure clustering channel is capable of large gains on individual seeds but not reliably.

What we take from this. The saturated clustering channel was not a defect holding the method back. Across the three arms the ordering is monotone in how active the clustering term is: clustering-only under cosine (0.477), full composite under cosine (0.537), full composite under the original construction (0.766). The more the clustering channel moves, the worse the retention. The honest reading is that the original formulation works *because* it is, in effect, a degree-variance regulariser, and that adding a live topological channel degrades it rather than improving it. This closes the open direction stated in the mechanism section and in Appendix A, in the negative.

We flag two limits on this conclusion. It is specific to catastrophic forgetting under LoRA on this model and task pair, and says nothing about the index used as a monitoring statistic in domains where the graph is built from correlations rather than weights. And the cosine construction is applied uniformly to all modules, whereas the original implementation branches on matrix shape (see Section 3.6), so for square modules the comparison differs in two respects rather than one.

B.13. First run: five-seed replication

Before the ten-seed experiment we ran a smaller five-seed version with the same protocol but with Omega-S at its *calibrated* strength (penalty set to a fixed fraction of the task gradient) rather than the swept target used above. We report it as a separate experiment because the Omega-S configuration differs; the two should be read as independent runs, not pooled.

Table 8: First run, five seeds, calibrated Omega-S. Retention as in Table 1. The direction matches the ten-seed result but is weaker and, against EWC, not separable at this seed count.

Seed	Weight decay (tuned)	EWC ($\lambda=10^3$)	Omega-S	Δ vs WD
42	48.94%	51.22%	50.00%	+1.1 pp
123	72.34%	77.78%	72.22%	−0.1 pp
456	56.10%	66.67%	66.15%	+10.1 pp
789	60.47%	65.91%	92.86%	+32.4 pp
1011	50.00%	39.13%	77.08%	+27.1 pp
Mean	57.57%	60.14%	71.66%	+14.1 pp

In this first run Omega-S beat tuned weight decay on 4 of 5 seeds (+14.1 pp mean). Against EWC the five seeds did not separate the two methods: Omega-S had the higher mean (71.7% vs. 60.1%) but EWC retained more on three of the five, with per-seed spread comparable to the gap. The ten-seed experiment, with Omega-S tuned rather than calibrated, is what resolves this: there Omega-S leads EWC on 8/10. Both runs agree on the robust conclusion (Omega-S over weight decay), and the larger run additionally separates it from EWC. We keep both tables so the reader can see the effect grow with sample size rather than shrink.

In both runs Omega-S ends with the highest absolute post-prose HumanEval of all arms (0.223 at ten seeds against 0.168 with no regulariser and 0.162 for weight decay). We report this as retained capability, not as plasticity, for the reason given in Section 3.2. All of this is obtained with zero architectural modification, zero access to previous domain data, and a training overhead of +1.5% (Section B.14).

B.14. Infrastructure Overhead: Single-GPU and FSDP

Table 9: Single GPU profiling (RTX 4090, Llama-3-8B + LoRA, bfloat16).

Scenario	VRAM Peak (GB)	Latency/step (ms)	Omega Cost (ms)
Baseline (LoRA)	21.98	251.7	0.0
Omega-S $K=1$	21.99	348.4	41.4
Omega-S $K=10$	21.99	261.1	3.5

VRAM overhead: +13 MB (+0.06%). At $K=10$, latency overhead is +3.7% with an isolated Omega cost of 3.5 ms per application.

Table 10: Two-GPU FSDP profiling ($2\times$ RTX 4090 PCIe, Llama-3-8B + LoRA).

Scenario	VRAM/GPU (GB)	Latency/step (ms)	Omega Cost (ms)
Baseline FSDP	17.18	14,343	0.0
Omega-S $K=1$	17.18	14,273	81.3
Omega-S $K=10$	17.18	14,565	8.1

The relative overhead of Omega-S at $K=10$ is +1.5%, lower than in the single-GPU setting, because LoRA adapter weights are not sharded by FSDP, meaning Omega-S computation requires **zero all-gather communication**. *The overhead does not scale with the number of GPUs.*

Note: the high absolute latency in Table 10 reflects PCIe interconnect limitations (no NVLink) in the experimental pod. Relative overhead figures are hardware-independent.

B.15. What Omega-S Is and Is Not

Omega-S is a structural regulariser: it acts on the weight connectivity graph $A = WW^\top$, not on the curvature of the loss landscape, so it is complementary to SAM (a full characterisation of their interaction is left to future work). We are, however, precise about *how* it acts. Its objective is topological by construction (built from $\text{Tr}(A^3)$, whose node-wise decomposition carries non-spectral information, Section 3), but the direct measurement of Section 3.3 shows that, as formulated, the penalty acts entirely through the degree-variance term: the clustering channel is saturated by the logistic map building A and does not move ($\Delta C \approx 0$ across all ten seeds). We therefore describe Omega-S, in its present form, as a *degree-variance regulariser with a topological objective whose clustering channel is currently inert*, rather than as a working topological method. Whether the effect is fully reducible to equalising weight-row norms is not settled: degree variance correlates $r \approx 0.6$ with row-norm variance here, leaving a substantial remainder, and a control swept on its own grid does not reproduce the effect: it fails to beat no regularisation reliably (6/10) while Omega-S exceeds it on 8/10 (one-sided $p = 0.055$). That is evidence against full reducibility, short of conventional significance at this seed count. The saturation is a design artefact of the $\sigma(|WW^\top|)$ construction, not an intrinsic limit of the approach; the reformulation in the Appendix would restore the clustering channel and let a genuinely topology-driven variant be evaluated.

B.16. Structured Sparsity as an Emergent Property

Omega-S alone does not produce structured sparsity (Table 12, $\lambda=0$ row). Structured sparsity emerges only in combination with group-lasso regularisation. In the MLP pruning experiments Omega-S shifts *which* neurons become structurally redundant while group-lasso provides the explicit pressure to collapse them to zero. We describe this as degree-structure-guided group

sparsity; given the mechanism measurement above, we avoid attributing it specifically to “topological centrality” and note that the marginal contribution of Omega-S over group-lasso alone in this pruning setting remains to be isolated.

B.17. Biological Analogy and Its Limits

The microbial network analogy is productive but imperfect. Biological scale-free networks emerge from evolutionary pressure over millions of generations; neural weight graphs emerge from gradient descent over thousands of steps. The analogy provides a structural intuition, not a mathematical equivalence: systems that process and route information efficiently tend to avoid extreme connectivity concentration. The empirical validation of this intuition constitutes the contribution of this paper; the biological framing provides motivation and vocabulary, while the mathematical and empirical results stand independently.

B.18. Broader Applications

Beyond structural pruning and sequential fine-tuning, Omega-S opens several promising research directions.

In **model quantisation**, weight monopolies often correlate with activation outliers, a primary bottleneck for sub-8-bit post-training quantisation that motivates outlier-handling mechanisms such as `LLM.int8()` [19] and SmoothQuant [20]. By redistributing weight density, Omega-S might yield networks that are inherently more quantisation-friendly. We flag this as a hypothesis rather than a finding: in preliminary probing at small scale we observed the expected reduction in degree variance but no corresponding improvement in post-quantisation accuracy, with weight decay performing comparably or better. Whether the effect appears at LLM scale, where outlier structure is far more pronounced, remains open.

In **Federated Learning**, applying a unified topological prior across decentralised nodes could mitigate structural divergence of local models trained on heterogeneous data, potentially improving weight aggregation stability. The absence of inter-GPU communication cost demonstrated in Section 4.5 suggests negligible overhead even in bandwidth-constrained settings.

In **Mixture of Experts (MoE)** architectures, the framework could penalise routing monopolies, preventing expert collapse and ensuring a more balanced token-to-expert assignment distribution.

The formulation extends beyond artificial neural networks. In **computational biology**, applying Omega-S constraints to graph neural networks predicting biological pathways could yield representations better aligned with the natural robustness of living systems. In **quantitative finance**, the penalty could regularise financial ML models to inherently penalise over-reliance on clustered economic nodes, yielding more robust portfolio optimisation and risk forecasting.

These interdisciplinary applications underscore the fundamental nature of topological regularisation across any domain where network structure dictates system resilience, a principle observed in global soil microbiomes [1,2,4] long before it was formalised in the context of artificial neural networks.

C. Declaration of LLM Usage

Large language models were used as writing and coding assistants during the preparation of this manuscript. Specifically, they were used to improve language clarity and presentation, to assist with coding and debugging for the numerical experiments, and to help identify and formalise related work. All technical ideas, theoretical results, experimental design and final interpretations were developed, verified and approved by the author.

D. Additional tables

Table 11: Per-seed retention, ten-seed experiment. Δ columns are Omega-S (log-ratio) minus each arm, in percentage points. Omega-S wins 8/10 against no regulariser, 9/10 against weight decay and 8/10 against EWC. The seeds it loses are informative in both cases: against EWC (123, 3033) they are the two where EWC posts its highest retention, and against the no-regulariser arm (123, 5055) they are the first and third highest of that column.

Seed	None	W. decay	EWC	Omega-S	Δ none	Δ WD	Δ EWC
42	66.7%	64.4%	68.3%	62.3%	-4.4	-2.2	-6.0
123	85.1%	69.2%	93.6%	87.8%	+2.7	+18.5	-5.9
456	69.6%	67.4%	77.1%	91.1%	+21.5	+23.7	+14.0
789	51.2%	58.1%	55.3%	105.6%	+54.3	+47.4	+50.2
1011	32.6%	44.4%	40.8%	70.8%	+38.2	+26.4	+30.0
2022	69.2%	69.6%	71.4%	82.5%	+13.3	+12.9	+11.1
3033	61.9%	64.4%	90.7%	83.7%	+21.8	+19.2	-7.0
4044	68.2%	69.6%	87.8%	85.5%	+17.3	+15.9	-2.3
5055	72.1%	63.6%	60.0%	74.1%	+2.0	+10.4	+14.1
6066	52.2%	47.9%	46.0%	97.6%	+45.4	+49.7	+51.6
Mean	62.9%	61.9%	69.1%	84.1%	+21.2	+22.2	+15.0

Table 12: Group-lasso λ sweep (Omega-S + GL combined).

GL λ	Accuracy	Sparsity	Dead Cols	Δ FLOPs	Structured
0.0 (Omega-S only)	95.45%	3.17%	0.00%	0.00%	$\times$
3×10^{-3}	95.52%	9.66%	5.65%	3.40%	$\checkmark$
1×10^{-2}	94.89%	48.22%	31.59%	26.32%	$\checkmark$

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A. A contrast-preserving reformulation of the clustering term

Section 3.3 showed that, as implemented, the clustering channel C is inert: the pseudo-adjacency $A = \sigma(|WW^\top|)$ compresses all affinities into $[0.5, 1)$, so A is near an all-ones matrix and the normalised clustering $\text{Tr}(A^3)/\|A\|_F^3$ sits at its ceiling (≈ 0.9997) irrespective of weight structure. The penalty can therefore only act through the degree-variance term Coex. This appendix sketches how to restore an informative clustering channel, so that a genuinely clustering-driven variant of Omega-S can be tested. We present it as an open direction for the community, not as a result of this paper.

The problem is the saturating map, not $\text{Tr}(A^3)$ itself. Any of the following keeps C responsive to structure:

(i) Temperature-scaled affinity. Replace $A = \sigma(|WW^\top|)$ with $A_\tau = \sigma(|WW^\top|/\tau - b)$, choosing the temperature τ and bias b so that the affinities span a usable range of $[0, 1]$ rather than collapsing to $[0.5, 1)$. Concretely, set b to the median of $|WW^\top|/\tau$ so that roughly half the entries fall below 0.5; then A_τ has genuine contrast and $\text{Tr}(A_\tau^3)$ varies with the actual triadic structure. The objective and its Hutchinson estimator are otherwise unchanged.

(ii) Thresholded graph. Form a sparse graph $A_\theta = \mathbf{1}\{|WW^\top| > \theta\}$ (or a smooth surrogate $\sigma(\kappa(|WW^\top| - \theta))$ with large κ), with θ a per-layer quantile of $|WW^\top|$. Clustering on a thresholded graph is the standard network-science definition and is not pinned near 1; its third moment then measures true triadic closure. The trade-off is that the hard threshold must be relaxed (via the smooth surrogate) to keep the penalty differentiable.

(iii) Rank-normalised affinity. Map $|WW^\top|$ through its empirical CDF so that A is uniform on $[0, 1]$ by construction. This removes scale and saturation effects entirely and makes C depend only on the *ordering* of affinities, i.e. on graph topology rather than magnitude.

Under any of these, one can define a clustering-only arm and a degree-variance-only arm (the row-norm / Coex control of the main text, re-tuned to remain a functional learner). One point must be settled before such an arm is run, and we flag it rather than resolve it: *nothing measured here determines the sign of the clustering term*. The objective as implemented minimises $\text{Tr}(A^3)$; the resilience index from which it derives [21] carries the clustering coefficient in its numerator and therefore rewards raising it; and the measurement of Section 3.3 does not arbitrate between the two, because C did not move at all. A clustering-only arm should accordingly test *both* directions, penalising $+\text{Tr}(A^3)/\|A\|_F^3$ and $-\text{Tr}(A^3)/\|A\|_F^3$, rather than assuming one. An earlier draft of this appendix specified only the second, which was an unexamined inheritance from the index rather than a considered choice. Running both, plus full Omega-S, on the ten-seed protocol would decompose the retention gain into a clustering component and a degree-variance component, and would settle whether the topological channel (inert here only because of σ) contributes once it is allowed to. We conjectured, in an earlier version of this appendix, that a contrast-preserving C would add retention on the seeds where the weight graph departs most from a degree-only description (the $\approx 40\%$ of degree variance not captured by row norms).

Outcome. That experiment has since been run and the conjecture does not hold. Section B.12 reports it in full: a row-wise cosine affinity removes the saturation, makes the clustering channel steerable during training, and then loses to the original formulation on all ten seeds ($p = 0.002$), with a clustering-only arm worse still. We leave this appendix in place because the diagnosis it

contains is correct and because the reasoning that led to the reformulation is what made the test possible; the conjecture at its close is what the evidence overturns.